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MIGHTY TRONICS®

PUMP INTERFACE FUNCTIONAL & ENGINEERING OVERVIEW

Version & Revision
V 1.0

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1. HISTORY OF VERSION

Date	Version	Amendments
26-09-2026	V1.0	Document initiation

2. ENGINEERING STATUS

This is an explanatory concept note, not a qualification report. Values derived from the original 4 A concept are recalculated for a 3 A continuous-current target. No 36 V / 3 A four-channel thermal or EMC test results were supplied.

3. WHAT IT IS INTENDED TO DO

Item	Revised / boundary
Channels	Four separately controlled DC pumps.
Power	External 5-36 V DC supply; up to 3 A continuous per channel, 12 A total.
Control	External 3.3 V or 5 V MCU PWM; 20-31 kHz; external 5 V logic supply.
Load and wiring	Inductive pump loads; cable cases up to 2 m in the charter, 4 m harness in a critical simulation.
Environment	Design study considers -10 to 50 deg C ambient; enclosure and final certification remain outside this phase.

4. HOW THE FUNCTIONS FIT TOGETHER

Power entry: receives and distributes pump power to four channels; reverse-polarity and supply transient protection must be rated for the actual current path. Control interface: conditions four PWM signals for the switching devices.

Pump drives: switch each pump independently. Transient control: flyback path, suppression, and RC damping limit switching stress.

Connections and service: labelled terminals and test points support assembly and diagnosis.

The external system supplies the pumps, 5 V logic, PWM generation, and cabling.

PWM duty cycle can adjust delivered motor power; actual speed and current depend on the pump and operating conditions.

5. ENGINEERING CALCULATIONS

5.1. CURRENT AND POWER ENVELOPE

Steady-state design current: $I_{\text{total}} = 4 \times 3 \text{ A} = 12 \text{ A}$. At 36 V, the upper electrical input envelope is $36 \text{ V} \times 12 \text{ A} = 432 \text{ W}$ at full on-time.

This is pump input power, not heat dissipated on the PCB. An external supply and the common input path must handle 12 A continuous plus validated startup margin. The earlier 6 A/channel startup target is not automatically retained; pump stall and startup measurements are needed.

5.2. MOSFET LOSSES: ILLUSTRATIVE ESTIMATES

For a hypothetical MOSFET $R_{\text{DS(on)}} = 20 \text{ milliohm}$ at the actual gate voltage and hot junction temperature:

$P_{\text{cond}} = I^2 R_{\text{D}}$. With $I = 3 \text{ A}$ and $D = 1$, $P_{\text{cond}} = 9 \times 0.020 = 0.18 \text{ W/channel}$, or 0.72 W for four.

If hot $R_{\text{DS(on)}}$ reaches 40 milliohm, the same estimate doubles to 0.36 W/channel. The selected device datasheet and actual gate drive must replace these assumed values.

A rough switching estimate is $P_{\text{sw}} = 0.5 \text{ V } I (t_{\text{r}} + t_{\text{f}}) f$. At 36 V, 3 A, 31 kHz and 100 ns **combined transitions**: 0.17 W/channel. At 300 ns: 0.50 W/channel.

Thus a preliminary MOSFET total might span about 0.35-0.86 W/channel using the two illustrative pairs above. It omits ringing, gate-drive loss and transient stress. Measure V_{DS} , I and switching times on the assembled board.

5.3. THE 15SQ100 IS A DESIGN CHECKPOINT, NOT A 12A PROOF

The 15SQ100 is rated 15 A average forward current at lead temperature 90 deg C in the referenced manufacturer data sheet, with 100 V repetitive reverse rating. The 15 A figure depends on heat removal and waveform. The supplied legacy schematic does not identify this diode; its installed position and exact manufacturer must be checked against the current PCB/BOM.

If one 15SQ100 is in series with the common supply, it could conduct the full 12 A continuously. Using the data sheet maximum V_{F} of 0.82 V specified at 15 A as an illustrative bound gives $12 \text{ A} \times 0.82 \text{ V} = 9.84 \text{ W}$ (not an exact 12 A prediction). An axial diode dissipating this power requires a real thermal analysis; 12 A should not be approved on nameplate current alone. If the diode is a freewheel device on one channel, its average current depends on that channel current and PWM off-time, rather than on four-channel input current. A TVS has yet another pulse-energy duty and must not be treated as a continuous-current diode.

6. MAKING THE DESIGN ROBUST

6.1. COPPER AND CONNECTOR CHECKS

For a 100 mm long, 2.0 mm wide, 70 micrometer copper segment at 20 deg C, $R = \rho L / (w t)$ with $\rho = 1.72 \times 10^{-8}$ ohm meter gives 12.3 milliohm. At 3 A: drop = 0.037 V and loss = 0.11 W. If the same geometry carries all 12 A: drop = 0.147 V and loss = 1.77 W. Copper resistivity rises with temperature; vias, neck-downs, terminals and inner-layer cooling can dominate. The 2 mm figure is an example for a branch, not an approval for the common feed. Size and test the actual stack-up and current return path, including connector ratings and wire gauge.

6.2. SWITCHING AND INDUCTIVE ENERGY

At 31 kHz the PWM period is $1/31,000 = 32.3$ microseconds. The concept simulations found significant drain ringing with cable parasitics and favoured a 22 ohm + 1 nF series RC snubber as a preliminary candidate. A 1nF capacitor cycled across 36 V has $0.5 C V^2 = 0.65$ microjoule of stored energy; multiplied by 31 kHz this is about 0.020 W for one such charge event per cycle. Actual resistor stress depends on the measured waveform and number of transitions. Its voltage rating, pulse rating and temperature need verification.

A freewheel path controls motor current when the switch opens. The snubber reduces ringing due to stray inductance; a supply TVS addresses supply transients. These parts serve different purposes. Measure worst-case MOSFET V_{DS} against its voltage rating at 36 V, with realistic pump, cable, current and supply. Layout should keep the switching loops short and the return paths continuous.

6.3. EVIDENCE REQUIRED BEFORE A ROBUSTNESS CLAIM

Question	Verification
Does every PWM channel switch correctly?	Logic-only, current-limited tests of input, buffer output and gate; resolve the observed abnormal levels first.
Can common and branch paths carry current?	36 V / 3 A per channel, all four active for one hour; log connector, diode, copper and MOSFET temperatures.
Do startup and fault currents remain safe?	Measure the intended pumps during start and stall; review fuses, diode waveform, supply and MOSFET safe operating area.
Are switching spikes controlled?	Scope V_{DS} , gate and current with the longest intended harness, synchronized and staggered PWM; tune snubber if required.
Is EMC behaviour acceptable?	Conducted and radiated pretests on final PCB and wiring; record setup and limits.

6.4.CONCLUSION

The 3 A/channel, 12 A total target is a sensible revised engineering envelope to investigate. It is not yet a demonstrated board rating. The decisive checks are the actual 15SQ100 current path and thermal mounting, common copper and connector sizing, MOSFET switching stress, pump startup behaviour, and four-channel prototype tests.

6.5.SOURCE BASIS

Project inputs: VS1 Project Charter - Pump Interface (31 Jan 2026) and VS2 Concept Engineering – Pump Interface (7 Feb 2026), both written for the earlier 4 A/channel target; revised target supplied 26 Sep 2026. Diode ratings: Rectron, 15SQ100 Schottky Barrier Rectifier, Rev. O, 2024-06, https://www.rectron.com/public/product_datasheets/15sq100.pdf. Assumed MOSFET resistance and transition times above are illustrative, not component guarantees.